

Soft Spaces Phase 3.2

OQS Theoretical Bridge

Projection-operator elimination, effective $P \rightarrow Q \rightarrow P$ feedback, and the degeneracy-closed interference law

Purpose. This note records the theoretical bridge between the Phase 3.2 mechanism and the projection-operator formalism presented by Breuer and Petruccione in Chapter 9, especially the Nakajima–Zwanzig construction on pp. 431–435. The purpose is not to claim that the two formalisms are identical, but to identify the common elimination architecture and to place the Soft Spaces interference law on a recognized open-quantum-systems foundation.

1. The projection-operator architecture in OQS

Breuer and Petruccione split the total density-operator dynamics into a relevant and an irrelevant part using complementary projection superoperators $\mathcal{P}$ and $Q = 1 - \mathcal{P}$. Their coupled equations have the schematic form

$$\frac{d}{dt}\mathcal{P}\rho(t) = \mathcal{P}\mathcal{L}\mathcal{P}\rho(t) + \mathcal{P}\mathcal{L}Q\rho(t), \quad (1)$$

$$\frac{d}{dt}Q\rho(t) = Q\mathcal{L}\mathcal{P}\rho(t) + Q\mathcal{L}Q\rho(t). \quad (2)$$

The Q equation is solved formally and substituted back into the $\mathcal{P}$ equation. In the Nakajima–Zwanzig equation this produces a memory term with kernel of the form

$$\mathcal{K}(t, s) \sim \mathcal{P}\mathcal{L}(t)\mathcal{G}(t, s)Q\mathcal{L}(s)\mathcal{P}, \quad (3)$$

where $\mathcal{G}(t, s)$ propagates the eliminated sector.

The structural message is simple: the relevant sector is influenced by a round trip

$$\mathcal{P} \longrightarrow Q \longrightarrow \text{propagation in } Q \longrightarrow \mathcal{P}.$$

This is the key OQS idea that is directly useful for Phase 3.2.

At second order, the kernel reduces schematically to a $\mathcal{P}\mathcal{L}Q\mathcal{L}\mathcal{P}$ structure. This is the same perturbative order at which the Soft Spaces effective P -space feedback appears.

2. The Soft Spaces energy-domain analogue

In Phase 3.2, P projects onto a selected degeneracy-closed eigenspace of the Hamiltonian and $Q = I - P$ onto its complement. Eliminating the Q sector produces the effective second-order object

$$\Sigma_E = PVQ(E - QHQ)^{-1}QVP \quad (4)$$

or, in the regularized implementation,

$$\Sigma_E = PVQ g_\eta(E - QHQ)QVP, \quad g_\eta(x) = \frac{x}{x^2 + \eta^2}. \quad (5)$$

The resolvent is the energy-domain counterpart of propagation in the eliminated sector. Schematically,

$$\mathcal{G}(t, s) \longleftrightarrow (E - QHQ)^{-1}. \quad (6)$$

The correspondence is structural rather than literal: Breuer–Petrucione work in Liouville space with superoperators, while Soft Spaces uses Hilbert-space projectors on Hamiltonian eigenspaces.

Important distinction. Breuer–Petrucione’s $\mathcal{P}$ and $\mathcal{Q}$ are projection *superoperators* acting on density operators. Our P and Q are ordinary Hilbert-space projectors. Their symbol $\Sigma(t)$ in the TCL construction is also not the same object as our Σ_E . The connection is the elimination mechanism, not an identity of notation or operator type.

3. Degeneracy-closed channel decomposition

The Phase 3.2 correction in v32.14 groups the eliminated sector into complete degenerate eigenspaces G , with projector P_G . The effective feedback decomposes as

$$\Sigma_E = \sum_G T_G, \quad T_G = g_\eta(E - E_G) PVP_GVP. \quad (7)$$

Each T_G is therefore one complete virtual return channel

$$P \xrightarrow{V} P_G \subset Q \xrightarrow{V} P,$$

weighted by its energy-domain propagation factor. Because P_G is the projector onto the entire degenerate subspace, this decomposition is invariant under arbitrary unitary rotations of basis vectors inside G .

4. The interference term emerges analytically

Using the Frobenius inner product,

$$\langle A, B \rangle_F = \text{Tr}(A^\dagger B), \quad (8)$$

we obtain

$$\|\Sigma_E\|_F^2 = \left\| \sum_G T_G \right\|_F^2 \quad (9)$$

$$= \sum_G \|T_G\|_F^2 + 2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K). \quad (10)$$

The first term measures the strengths of the virtual channels separately. The second term measures their mutual constructive or destructive operator interference.

This motivates the basis-invariant normalized interference parameter

$$A_E^{(G)} = \frac{2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K)}{\left(\sum_G \|T_G\|_F \right)^2}. \quad (11)$$

The associated coherence factor is

$$C_E^{(G)} = \frac{\|\sum_G T_G\|_F}{\sum_G \|T_G\|_F}. \quad (12)$$

Thus $C_E^{(G)}$ measures how coherently the virtual return channels combine, while $A_E^{(G)}$ isolates the normalized cross-channel contribution explicitly.

5. What v32.14 adds to the OQS picture

The OQS projection formalism explains why eliminating an irrelevant sector generates an effective feedback kernel on the relevant sector. Phase 3.2 adds a more specific finite-dimensional observation: the local hotspot signal is associated not merely with the magnitude of the eliminated-sector feedback, but with the *interference among its channel contributions*.

For v32.14, using the degeneracy-closed definition above, the interference association survives across the 12 frozen seeds. For $A_E^{(G)}$, the median seedwise Spearman correlations with hotspot prominence are approximately

$$\rho_{\text{med}} = 0.2605 \quad (\text{dephasing}), \quad \rho_{\text{med}} = 0.2843 \quad (\text{transverse}), \quad (13)$$

with positive sign in 10/12 seeds for both families. The coherence factor $C_E^{(G)}$ is positive in 12/12 seeds for both perturbation families, with median seedwise correlations approximately 0.2500 and 0.5194, respectively.

Phase 3.2 working interpretation.

Hotspot prominence increases statistically when the virtual $P \rightarrow Q_G \rightarrow P$ channels combine more coherently.

The OQS projection framework supplies the elimination architecture; the Soft Spaces result identifies a candidate interference observable inside that architecture.

6. Consequence for the next step

The theoretical bridge now justifies the planned robustness study. v32.15 should no longer be described merely as a generic parameter scan. Its sharper purpose is:

Test whether the OQS-motivated, degeneracy-closed interference law remains stable under controlled model variations.

The principal quantities to preserve are $A_E^{(G)}$ and $C_E^{(G)}$, while varying Hamiltonian term count, seed ensemble, and selected regularization settings one factor at a time.

If the sign and strength of the hotspot association persist, the result moves from a frozen-model mechanism to a candidate structural law. If they do not, the present result should be retained as a model-specific but theoretically interpretable phenomenon.

Source basis. Breuer and Petruccione, *The Theory of Open Quantum Systems*, Chapter 9, especially the projection-operator discussion on pp. 431–435 and the equations surrounding the Nakajima–Zwanzig kernel and its second-order approximation. This note uses those pages as theoretical grounding and deliberately distinguishes their Liouville-space superoperators from the Hilbert-space projectors used in Soft Spaces.